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Fair Division

Selected Topics in Algorithms and Optimization (Modul V5C4)

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Lecture 1: Cake-cutting

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These notes are partly based on lecture notes by Masoud Seddighin.

Outline of this session. Introduction to fair division, cake-cutting, moving-knife procedures, and the Robertson–Webb model.

Computational Social Choice Theory

Social choice theory studies how to make a *collective* decision from *individual* preferences. Three central areas are:

- **Voting**
- **Resource allocation**
- **Stable matchings**

The field is approached from two complementary perspectives:

- **Economists and politicians:** Does there exist a **social choice** mechanism with the desired economic properties?
- **Mathematicians / computer scientists:** How can we (efficiently) **compute** such a mechanism?

In this course, we focus on **resource allocation**.

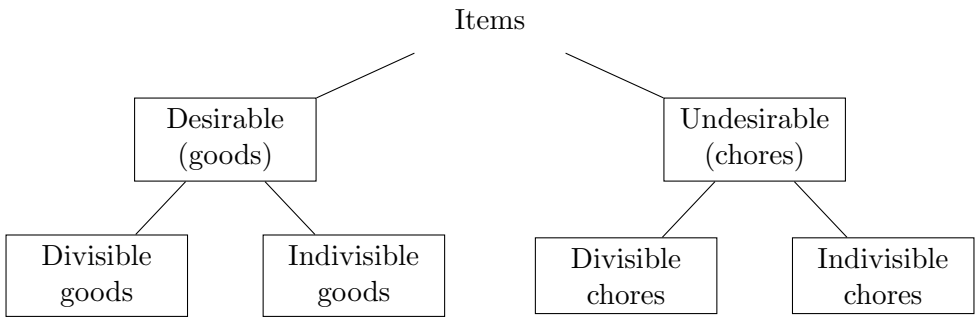
Fair Division

Definition 1. Fair division is the problem of dividing **items** among **agents** in a **fair** manner.

Applications: partnership dissolution, divorce settlements, budget allocation, land division, etc.

Types of Items

Items can be *desirable* (goods, e.g. assets in a divorce settlement) or *undesirable* (chores, e.g. household chores). Each type can, in turn, be divisible or indivisible:



Two Central Questions

- What is **fairness as a concept**?
- How do we **compute a fair allocation**?

Goal: to **divide** the resource among the agents in a **fair** manner.

Cake Cutting

Fair division of indivisible goods is also known as "cake cutting". For now, our focus is on the cake cutting problem.

Formal Definition

In the cake cutting problem, we have one cake (one resource) that must be divided among a number of individuals.

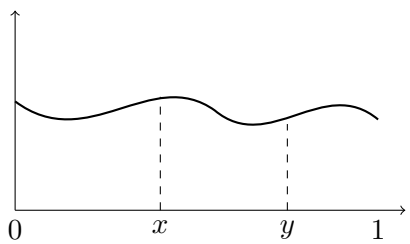
- **Cake C:** we model the cake as the interval between zero and one, $C = [0, 1]$.
- **Set of agents:** $N = [n] = \{1, 2, \ldots, n\}$, the agents among whom the cake must be divided.
- **Valuation function of player i:** v_i. For every subinterval I of $[0, 1]$, $v_i(I)$ specifies the value of that subinterval.

We assume that the valuation functions have the following properties:

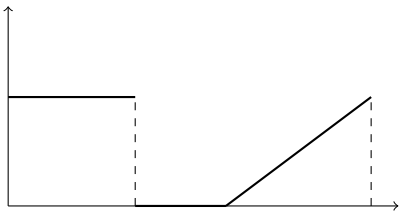
1. **Normalization:** the valuation functions are normalized so that $\forall i: \ v_i(C) = 1$.
2. **Divisibility:** for every interval $[x, y]$ and every $0 \leq \lambda \leq 1$, there exists a point $z \in [x, y]$ such that
$$v_i([x, z]) = \lambda \, v_i([x, y]).$$
The second condition implies that the valuation functions are *non-atomic* and that intervals of length zero have value zero. Hence, in our discussions, there is no difference between "]" and ")".
3. **Non-negativity:** $\forall i, \ \forall I: \ v_i(I) \geq 0$.

Definition 2. A **piece** of the cake is a set of finitely many disjoint intervals of the cake. For a piece A, $|A|$ is the number of intervals in A. We also define
$$v_i(A) = \sum_{I \in A} v_i(I).$$

For convenience, we may assume that each individual has a density function v_i defined on $[0, 1]$, so that
$$v_i([x, y]) = \int_x^y v_i(t) \, dt.$$
The function differs between individuals.



However, v_i need not be continuous; some parts of the cake may have value zero, for example (piecewise integral):



Definition 3. We denote an **allocation** of the cake to the individuals by $A = (A_1, A_2, \ldots, A_n)$, where A_i is the piece (possibly several intervals) allocated to individual i.

Fairness Criteria

The goal of the problem is the fair division (allocation) of the cake among the individuals. But first, we must specify what *fair* means. Many fairness criteria have been introduced; we present three of them here:

1. **Proportional:** an allocation $A = (A_1, \ldots, A_n)$ is proportional if every individual feels that they have received at least $\frac{1}{n}$ of the cake:
$$\forall i: \ v_i(A_i) \geq \tfrac{1}{n}.$$
2. **Envy-free:** everyone prefers their own share to the shares of others:
$$\forall i, j: \ v_i(A_i) \geq v_i(A_j).$$
3. **Equitable:** everyone's level of satisfaction with their own share is the same:
$$\forall i, j: \ v_i(A_i) = v_j(A_j).$$

Additional Properties

Besides fairness, we may want the division to have other properties as well:

1. **Contiguous:** each individual receives exactly one interval: $\forall i: \ |A_i| = 1$.
2. **Pareto optimal:** there is no allocation A' in which nobody is worse off and at least one individual is strictly better off:
$$\nexists A': \quad \forall i: \ v_i(A'_i) \geq v_i(A_i) \quad \text{and} \quad \exists j: \ v_j(A'_j) > v_j(A_j).$$
3. **Non-wasteful:** the whole cake is allocated.
4. **Truthful:** if agents were asked to report their valuations, they would have no incentive to lie.

This is the overall definition of the fair allocation problem and of our goals in it.

Theorem 1. *In non-wasteful allocations, we have:*

- *EF $\Rightarrow$ proportionality.*
- *For 2 agents: EF $\iff$ proportionality.*
- *For $n > 2$ agents: proportionality $\not\Rightarrow$ EF.*

Proof. By additivity and normalization, for every agent i,

$$\sum_{j=1}^{n} v_i(A_j) = v_i(C) = 1. \tag{1}$$

EF $\Rightarrow$ proportionality. Let A be envy-free and fix an agent i. By envy-freeness, $v_i(A_i) \geq v_i(A_j)$ for all j. Summing over all $j \in N$ and using (1), we get

$$n\, v_i(A_i) \;\geq\; \sum_{j=1}^{n} v_i(A_j) \;=\; 1,$$

hence $v_i(A_i) \geq \frac{1}{n}$. Since i was arbitrary, A is proportional. This also proves both the "$\Rightarrow$" direction for 2 agents.

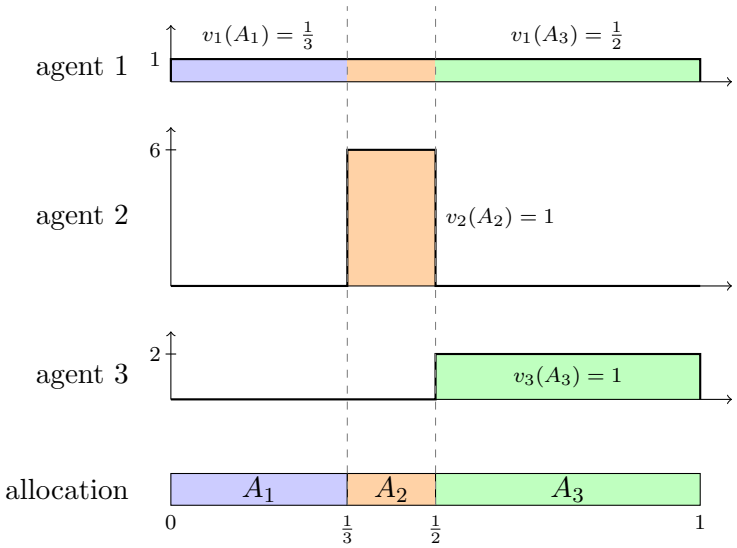
Proportionality $\Rightarrow$ EF for $n = 2$. Let $A = (A_1, A_2)$ be proportional, and let $i \in \{1, 2\}$ and j be the other agent. By (1) and proportionality,

$$v_i(A_j) \;=\; 1 - v_i(A_i) \;\leq\; 1 - \tfrac{1}{2} \;=\; \tfrac{1}{2} \;\leq\; v_i(A_i).$$

Hence no agent envies the other, so A is envy-free.

The converse fails for $n > 2$. Take $n = 3$ (for larger n the example extends in the same way). Let agent 1 have the uniform density on $[0, 1]$, agent 2 the uniform density on $[\frac{1}{3}, \frac{1}{2}]$, and agent 3 the uniform density on $[\frac{1}{2}, 1]$. Consider

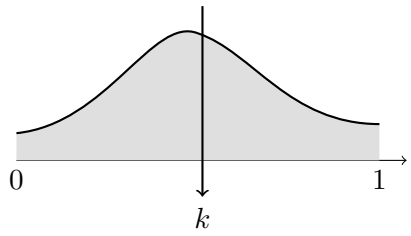
$$A_1 = [0, \tfrac{1}{3}], \qquad A_2 = [\tfrac{1}{3}, \tfrac{1}{2}], \qquad A_3 = [\tfrac{1}{2}, 1].$$



Then $v_1(A_1) = \frac{1}{3}$ and $v_2(A_2) = v_3(A_3) = 1$, so A is proportional. However, $v_1(A_3) = \frac{1}{2} > \frac{1}{3} = v_1(A_1)$, so agent 1 envies agent 3 and A is not envy-free. □

Algorithms

Example 1 (Dividing a cake between 2 agents: *Cut & Choose*)**.** This method is very old (Genesis 13). First, we ask agent 1 to divide the cake into two equal parts (according to their own valuation); then agent 2 chooses the piece that is more valuable to them, and the remaining piece goes to agent 1.



$v_1([0,k]) = \frac{1}{2} = v_1([k,1])$

Properties:

Proportional: ✓	Envy-free: ✓	Equitable: ✗
Contiguous: ✓	Pareto optimal: ✗	

Example 2 (Moving-knife procedure, Dubins & Spanier, 1961)**.** A procedure for proportional division among n agents.

A referee moves a knife from the beginning of the cake (point 0) to the right. This continues until one agent says "stop", meaning that the part to the left of the knife is worth $\frac{1}{n}$ to them. We cut that piece, give it to that agent, and continue the process with the remaining players.

Properties:

Proportional: ✓	Envy-free: ✗	Equitable: ✗
Contiguous: ✓	Pareto optimal: ✗	

A fundamental problem with the previous procedures is their *continuous* nature. Basically, in computer science continuous movements cannot be simulated without a loss of precision.

The Robertson–Webb model (1998)

In this model we may use only two types of query. That is, to obtain information about the agents' valuation functions, we can use only the following two queries:

- $\text{Cut}_i(x, \alpha)$: ask person i to specify the point y of the cake such that

$$v_i([x,y]) = \alpha,$$

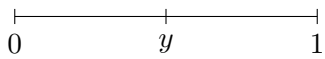
  where y lies to the right of x. (Note: this query does *not* actually cut the cake!)

- $\text{Eval}_i(x, y)$: ask person i to specify the value of the interval $[x, y]$, i.e. what $v_i([x,y])$ is equal to.

A cake-cutting algorithm that uses only these two types of query is called a *protocol*.

Example 3 (Implementing cut & choose in the Robertson–Webb model)**.**

1. $\text{Cut}_1(0, \frac{1}{2}) = y$
2. $\text{Eval}_2(0, y)$



With these two queries, cut & choose can be simulated completely.

Example 4 (Implementing the Dubins–Spanier moving-knife procedure in the Robertson–Webb model)**.**

1. First, we make n queries of the form $\text{Cut}_i(0, \frac{1}{n})$. Let y be the smallest value among the answers.
2. We give the interval $[0, y]$ to the person whose answer was y. In the next step we make $n-1$ queries of the form $\text{Cut}_i(y, \frac{1}{n}), \ldots$ We continue in this way until everyone has received a share.

Number of queries:

$$n + (n-1) + (n-2) + \cdots + 2 = O(n^2).$$

Can we obtain a proportional allocation with fewer queries?

Example 5 (Even–Paz algorithm, 1984)**.** Assume n is a power of 2. A referee moves a knife from the beginning of the cake to the right. This continues until $\frac{n}{2}$ of the individuals shout "stop", meaning that the part to the left of the knife is worth $\frac{1}{2}$ of the cake to them. We halve the cake at this point and solve the problem recursively for the two sides.

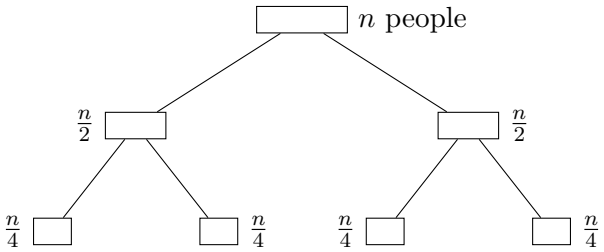
Properties:

Proportional: ✓	Envy-free: ✗	Equitable: ✗
Contiguous: ✓	Pareto optimal: ✗	

Example 6 (Implementing the Even–Paz moving-knife procedure in the Robertson–Webb model)**.**

1. $n \times \text{Cut}_i(0, \frac{1}{2}) \rightsquigarrow y$
2. $\frac{n}{2} \times \text{Cut}_i(0, \frac{1}{4}), \qquad \frac{n}{2} \times \text{Cut}_i(y, \frac{1}{4})$

⋮



Number of queries:

$$T(n) = 2\,T(n/2) + n \;\longrightarrow\; T(n) = O(n \log n).$$

So the Even–Paz procedure is better than the Dubins–Spanier procedure in terms of the number of queries. The question that now arises is: can a proportional division be achieved with even fewer queries?

Theorem 2 (Edmonds & Pruhs 2006)**.** *Every protocol for proportional cake division requires* $\Omega(n \log n)$ *queries.*

(SODA'06: *Cake cutting really is not a piece of cake.*)

Exercises

Question 1. Give a moving-knife procedure that outputs an equitable allocation for two agents.
Question 2. Give a Robertson–Webb protocol that outputs a proportional allocation using $O(n \log n)$ queries for arbitrary n (not necessarily a power of 2).
Question 3. For any arbitrary $n > 3$, prove that proportionality does not imply envy-freeness.